

Question	Answer	Marks
3(a)	force is proportional to extension	B1
3(b)(i)	length = 37 m	A1
3(b)(ii)	spring constant = F / x	C1
	$= \text{e.g. } 1800 / (120 - 37)$ $= 22 \text{ N m}^{-1}$	A1
3(c)(i)	$((\Delta)E) = mg(\Delta)h$	C1
	$= 64 \times 9.81 \times 120 = 75\,000 \text{ J}$ or 75 kJ	A1
3(c)(ii)	$(E =) \frac{1}{2}Fx$ or $(E =) \frac{1}{2}kx^2$ or $(E =) \frac{1}{2}F^2 / k$ or $(E =)$ area under graph	C1
	$(E =) \frac{1}{2} \times 1800 \times (120 - 37) = 75\,000 \text{ J}$ or 75 kJ or $(E =) \frac{1}{2} \times 21.7 \times (120 - 37)^2 = 75\,000 \text{ J}$ or 75 kJ	A1
3(d)	(all) gravitational potential energy has been converted / equal to elastic potential energy (so no kinetic energy)	B1
	kinetic energy is zero so speed is zero	B1

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